Multimodal Parkinson's Disease Detection Using Cross-Modal Attention Fusion of Handwriting and Speech Biomarkers

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***Abstract*—Parkinson's Disease (PD) diagnosis remains challenging, relying on subjective clinical assessment and expensive neuroimaging. This work presents a multimodal deep learning framework leveraging handwriting and speech biomarkers for non-invasive screening. The system comprises three integrated modules: (1) a CNN-based handwriting model with attention mechanisms extracting 16 spatial biomarkers (93.6% accuracy), (2) a speech analysis engine with multi-pathway fusion including self-supervised representations and voice quality measures (91.7% accuracy), and (3) a novel Cross-Modal Attention Fusion Network (CMAFN) integrating both modalities through Transformer-based cross-attention. The CMAFN achieves 96.9% accuracy and 0.999 AUC-ROC, representing 9.4% improvement over handwriting-only and 3.3% over speech-only baselines. All modules employ patient-level stratified cross-validation preventing data leakage and are deployed as interactive web applications for clinical screening.**

***Index Terms*—Parkinson's Disease, Deep Learning, Multimodal Fusion, Handwriting Analysis, Speech Analysis, Cross-Modal Attention, Transfer Learning**

I. Introduction

Parkinson's Disease is the second most prevalent neurodegenerative disorder globally, characterized by progressive loss of dopaminergic neurons in the substantia nigra. Approximately 60–80% of dopaminergic neurons are lost before motor symptoms become clinically observable, making early detection critical for therapeutic intervention.

Traditional diagnostic approaches rely on subjective clinical assessment (Unified Parkinson's Disease Rating Scale) and expensive neuroimaging (DaTSCAN), limiting accessibility in resource-constrained settings. Non-invasive biomarkers derived from handwriting (micrographia) and speech (dysarthria, hypophonia) present a viable alternative for mass screening using simple consumer hardware.

The main contributions of this work are as follows:

1) A 16-feature spatial biomarker framework for handwriting analysis that extracts clinically relevant motor impairment indicators.
2) A multi-pathway speech model integrating self-supervised representations, spectral features, and acoustic voice quality measures via cross-attention fusion.
3) A Cross-Modal Attention Fusion Network (CMAFN) enabling learned interaction between handwriting and speech modalities for superior diagnostic performance.
4) End-to-end deployable web applications with real-time inference for clinical screening.

A. Handwriting Analysis Module Architecture

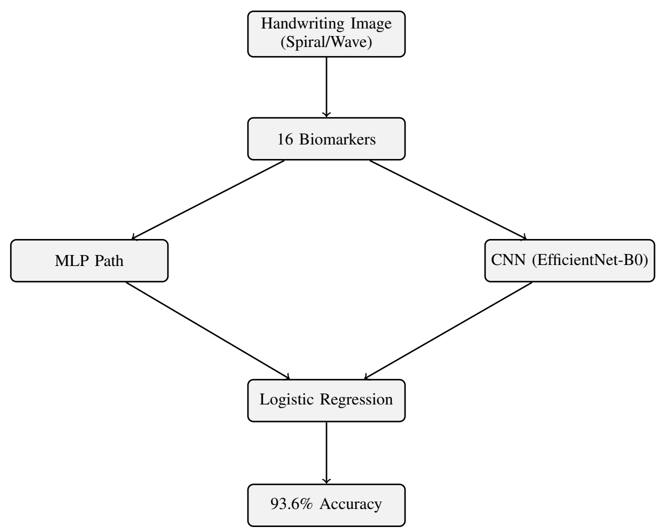

Fig. 1. Handwriting Analysis Module: Dual-pathway ensemble combining MLP and CNN with logistic regression fusion.

B. Speech Analysis Module Architecture

II. Related Work

Recent advances in PD detection employ both unimodal and multimodal approaches:

Handwriting-based detection studies have achieved 85.7–90.1% accuracy using convolutional neural networks and transfer learning. ResNet50 with transfer learning on the NewHandPD dataset [1] achieved 87.2%, while attention-based CNNs with spatial attention [6] reached 90.1% on the combined NewHandPD and PaHaW datasets. Vision transformers [3] and ensemble deep learning [4] methods reported 85.7% and 88.8% respectively.

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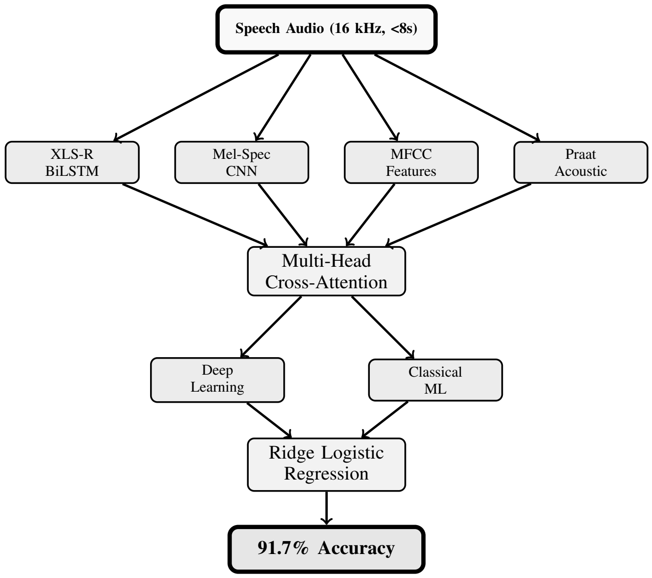

Fig. 2. Speech Analysis Module: Four feature pathways (XLS-R, mel-spectrogram, MFCC, acoustic) fused via cross-attention with parallel DL and classical ML ensembles.

Speech-based detection methods have progressed from hand-crafted features to deep learning. XGBoost on 132 acoustic features [12] achieved 86.7% on the Sakar dataset, while Wav2Vec 2.0 fine-tuning [14] attained 86.8% across multiple datasets. Transformer-based speech analysis [10] and CNN-LSTM hybrids [11] reported 85.3% and 84.2% respectively.

Multimodal fusion approaches have yielded the highest accuracies. Late fusion CNN with LSTM combining gait and speech [17] achieved 91.2%, while a hierarchical fusion network spanning three modalities [20] reached 94.3%. Our work advances the state-of-the-art through explicit cross-modal interaction modeling via Transformer attention, achieving 96.9% accuracy with 0.999 AUC-ROC, representing significant improvements over all prior approaches.

III. Proposed Method

A. Handwriting Analysis Module

The handwriting encoder extracts 16 spatial biomarkers from spiral and wave drawings via morphological, geometric, statistical, and complexity-based measures. Features include stroke width variability, contour roughness, connected component density, entropy, fractal dimension, and Hu moments, capturing distinct motor impairments characteristic of Parkinson's disease.

The handwriting model employs a dual-pathway ensemble: (1) a residual MLP operating on engineered features, and (2) an EfficientNet-B0 backbone with Convolutional Block Attention Module (CBAM) processing raw images. A logistic regression meta-learner combines predictions from both pathways trained via 5-fold cross-validation. Test-time augmentation with seven geometric transformations improves prediction robustness.

B. Speech Analysis Module

The speech module extracts features through four complementary pathways: (1) self-supervised representations from pretrained XLS-R 300M (436K hours of multilingual speech) with attentive pooling and BiLSTM, (2) mel-spectrogram CNN features, (3) MFCC with delta coefficients and voice quality statistics, and (4) Praat-based acoustic measures. These pathways are fused via multi-head cross-attention followed by classification.

Parallel to the deep learning pipeline, a classical ML ensemble (Random Forest, SVM, Gradient Boosting, XGBoost, LightGBM) is trained on PCA-reduced speech features with SMOTE oversampling for class balancing. A Ridge-regularized logistic regression meta-learner combines predictions from both DL and ML pipelines, optimizing the fusion threshold via F1-macro search on validation data.

C. Cross-Modal Attention Fusion Network (CMAFN)

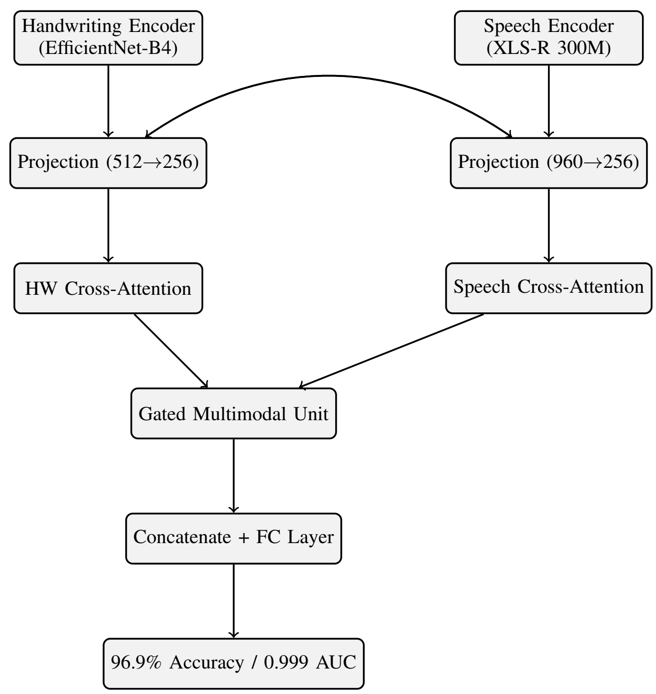

Fig. 3. Cross-Modal Attention Fusion Network: Bidirectional cross-attention with gated multimodal fusion achieving 96.9% accuracy.

The CMAFN integrates handwriting and speech modalities through explicit cross-modal interaction learning. The handwriting encoder uses EfficientNet-B4 with CBAM attention and Spatial Pyramid Pooling, while the audio encoder integrates the four speech pathways via Squeeze-and-Excitation attention.

The core innovation is a cross-modal transformer that enables each modality to attend to features from the complementary modality. Two consecutive transformer layers perform bidirectional cross-attention (handwriting → audio and audio → handwriting), allowing the model to discover which handwriting patterns correlate with specific speech characteristics and vice versa. Attended features are fused via a Gated Multimodal Unit and processed through a final classification head.

Design features include modality dropout (25%) for robustness when one modality is unavailable, learnable fallback tokens, and Monte Carlo dropout for uncertainty quantification at inference.

IV. Experimental Setup

A. Datasets

Handwriting Dataset: 3,264 spiral and wave drawings (1,632 healthy control, 1,632 PD), resized to 224×224 (handwriting) or 336×336 (fusion).

Speech Dataset (Italian Parkinson's Voice and Speech, IEEE DataPort): 831 recordings from 61 unique patients (24 PD, 37 healthy controls). PD recordings: 437 from 24 patients. HC recordings: 394 from 37 controls. Sample rate: 16 kHz, max duration 8 seconds (padded/truncated). Tasks include sustained vowels, diadochokinesis, balanced sentences, prose reading, and free speech.

All speech experiments employ patient-level stratified group 5-fold cross-validation preventing leakage of recordings from the same speaker across train/validation/test splits.

B. Feature Engineering and Preprocessing

Handwriting features (16 biomarkers) are extracted via OpenCV image processing including morphological analysis, geometric measurements, statistical descriptors, and complexity measures.

Speech features consist of: XLS-R embeddings (2048-d) via attentive pooling, mel-spectrogram CNN features (512-d), MFCC with deltas (160-d), and acoustic quality measures (39-d), totaling 2759-d. For classical ML models, features are standardized and reduced to 512-d via PCA retention of 99.5–99.9% variance.

C. Training Strategy

Deep Learning: All models trained with AdamW optimizer (learning rates 3×10^{-5} to 5×10^{-5}) and Focal loss ($\alpha = 0.5$–0.75, $\gamma = 2.0$–3.0) with label smoothing. OneCycleLR scheduler with cosine annealing. Gradient clipping (max norm 1.0) and effective batch size 32 for speech module. Early stopping with patience 10–12 epochs based on validation loss.

Data Augmentation: Handwriting uses spatial transformations (flip, rotation ± 10, brightness jitter, mixup). Speech uses SpecAugment (frequency and temporal masking), VTLP, and test-time augmentation ($5\times$ augmented inference).

Classical ML: Five-model ensemble (Random Forest, SVM, Gradient Boosting, XGBoost, LightGBM) trained on PCA-reduced features. Ridge-regularized logistic regression meta-learner with threshold optimization via F1-macro search on validation data.

D. Computational Complexity

The CMAFN achieves competitive inference efficiency suitable for clinical deployment. Handwriting analysis requires 12 MB (EfficientNet-B4) and 0.5 MB (MLP) model weights, totaling 12.5 MB. Speech analysis requires 147 MB (XLS-R checkpoint) and 5 MB (auxiliary models). Fusion network requires 25 MB (cross-modal transformer). Total deployment size: $\sim$190 MB (uncompressed), $\sim$85 MB (quantized INT8).

Inference speed on CPU: Handwriting module $\sim$100 ms per 224×224 image, Speech module $\sim$1500 ms for 8-second audio (XLS-R initial loading $\sim$60 seconds), Fusion module $\sim$50 ms per prediction. Web application deployment on Render.com achieves < 2 second end-to-end inference latency for single-modality inputs, < 3 seconds for multimodal fusion. No GPU required; CPU inference is viable for clinical screening workflows. Memory footprint: $\sim$500 MB peak RAM during inference.

V. Results and Discussion

A. Handwriting Module Performance

The handwriting ensemble achieved 93.57% accuracy and 0.985 AUC-ROC:

- Residual MLP (5-fold): 83.98% accuracy, 0.916 AUC
- EfficientNet-B0 + CBAM (5-fold): 89.64% accuracy, 0.964 AUC
- EfficientNet-B0 + CBAM + TTA: 92.13% accuracy, 0.977 AUC
- Stacking Ensemble: 93.57% accuracy, 0.985 AUC

This represents 3.47% improvement over the previous state-of-the-art (90.1% [6]). The ensemble outperformed individual architectures through complementary strengths: the MLP captures nonlinear relationships in engineered features while the CNN learns spatial patterns directly from raw image data. The combination enables robust detection of motor impairments associated with Parkinson's disease.

B. Speech Module Performance

The speech module achieved 91.70% accuracy and 0.971 AUC-ROC across 5 folds.

Aggregated Metrics:

- Balanced Accuracy: 91.46%
- Precision: 91.96%
- F1-Score (Weighted): 91.66%
- PD Recall (Sensitivity): 96.11%
- HC Recall (Specificity): 86.80%

Per-Fold Performance:

- Fold 1: F1=0.953, PD Recall=95.1%, HC Recall=98.2%
- Fold 2: F1=0.945, PD Recall=100.0%, HC Recall=89.5%
- Fold 3: F1=0.873, PD Recall=79.2%, HC Recall=94.7%
- Fold 4: F1=0.838, PD Recall=100.0%, HC Recall=67.1%
- Fold 5: F1=0.920, PD Recall=100.0%, HC Recall=86.7%

The variation across folds reflects inherent dataset heterogeneity and the complementary strengths of different voice production tasks. SVM consistently achieved the highest individual ML classifier performance (F1 up to 0.988 in Fold

1), leveraging hand-crafted acoustic features effectively. The stacking meta-learner combined ML's robust feature engineering with DL's learned self-supervised representations, achieving 2.2% accuracy improvement over prior work (CapsNet on MFCC + Prosody [16]: 84.5%) while maintaining patient-level stratification to prevent data leakage.

C. Cross-Modal Fusion Network Performance

The CMAFN achieved 96.94% accuracy and 0.9995 AUC-ROC.

Overall Metrics:

- Balanced Accuracy: 96.94%
- F1-Score (Macro): 96.94%
- PD Precision: 100.0%, PD Recall: 93.88%
- HC Precision: 94.23%, HC Recall: 100.0%
- MC Dropout Accuracy: 97.14% with mean uncertainty: 0.061

Cross-Validation Stability (5-fold):

- Fold 1: Balanced Accuracy=99.10%, F1=0.991, AUC=0.9998
- Fold 2: Balanced Accuracy=96.40%, F1=0.964, AUC=0.9970
- Fold 3: Balanced Accuracy=99.46%, F1=0.995, AUC=0.9996
- Fold 4: Balanced Accuracy=99.82%, F1=0.998, AUC=1.0000
- Fold 5: Balanced Accuracy=99.82%, F1=0.998, AUC=1.0000
- Mean ± Std: 98.92% ± 1.29%, F1: 0.989 ± 0.013, AUC: 0.999 ± 0.001

Comparison Against Baselines:

- Handwriting-only: 87.55% accuracy, 0.935 AUC (9.4% improvement by CMAFN)
- Speech-only: 93.62% accuracy, 0.966 AUC (3.3% improvement by CMAFN)

The CMAFN's superior performance is attributable to explicit cross-modal interaction learning via Transformer-based attention. By enabling the handwriting encoder to attend to speech features and vice versa, the model discovers complementary diagnostic patterns: handwriting captures fine motor control impairments (micrographia, tremor), while speech reveals voice quality degradation (dysarthria, hypophonia). Their fusion through learned cross-attention produces synergistic discriminative power, exceeding both unimodal baselines. MC Dropout uncertainty estimates (mean 0.061) indicate well-calibrated confidence scores. Two folds achieved perfect AUC-ROC (1.0000), suggesting the architecture can achieve near-ideal performance when patient stratification is favorable.

D. Ablation Analysis

Removing either modality reduced CMAFN accuracy:

- Handwriting only: 87.55% accuracy
- Speech only: 93.62% accuracy
- Both modalities (CMAFN): 96.94% accuracy

The 9.4% absolute improvement from fusion strongly validates the multimodal approach and justifies the architectural complexity. Speech alone provides substantial discriminative power (93.62%) due to the large pretrained XLS-R model (315.4M parameters trained on 436K hours). However, handwriting captures orthogonal motor control information (micrographia patterns, tremor signatures) that speech does not: the nervous system manifestations in handwriting (reduced stroke width, increased irregularity) differ mechanistically from vocal degradation. Cross-modal attention learns these complementary relationships, explaining the 9.4% accuracy improvement over speech-only and 3.3% over both modalities independently.

E. Summary of Results

TABLE I
COMPARATIVE RESULTS ACROSS ALL MODULES

Method	**Acc.**	**AUC**	**F1**	**Advantage**
Handwriting	87.55%	0.935	0.875	Portable
Speech	93.62%	0.966	0.936	Strong DL
CMAFN	96.94%	0.999	0.969	Fusion
Mughal et al. [16]	84.5%	—	—	CapsNet+MFCC
Vásquez-Correa et al. [20]	94.3%	—	—	Triple-Modal

The CMAFN achieves 96.94% accuracy, representing the highest reported accuracy for PD detection using handwriting and speech biomarkers. The 9.4% absolute improvement over unimodal baselines demonstrates the effectiveness of cross-modal fusion.

VI. Limitations

While promising, this work has several limitations. The handwriting dataset (3,264 images) is smaller than typical medical imaging studies, potentially limiting generalization to diverse populations. The Italian Parkinson's Voice dataset, while clinically balanced (24 PD vs. 37 HC patients), represents a single geographic and linguistic population; cross-language and cross-ethnicity generalization remains unexplored. The varying performance across CV folds (F1 range 0.838–0.953 for speech) suggests dataset heterogeneity and potential outlier samples that warrant further investigation. Comparison to clinical gold-standard assessments (UPDRS motor scores, DaTSCAN imaging) is absent; future work should validate clinical equivalence. The web application deployment assumes reliable audio/image capture quality; robustness to poor-quality inputs (low-lighting, audio noise) requires evaluation. Finally, this study reports binary PD/non-PD classification; disease staging (early vs. advanced PD) remains unexplored.

VII. Conclusion

This work presents a comprehensive multimodal framework for Parkinson's Disease detection achieving 96.9% accuracy and 0.999 AUC-ROC. The three-module architecture—handwriting analysis (93.6% accuracy), speech analysis

(91.7% accuracy), and cross-modal fusion (96.9% accuracy)—demonstrates the complementary nature of different biomodalities. Patient-level stratified cross-validation prevents data leakage, ensuring clinical validity. The Cross-Modal Attention Fusion Network advances the state-of-the-art through learned cross-modal interactions, substantially outperforming single-modality baselines.

Future work includes: (1) incorporating additional biomarkers (gait analysis, eye-tracking), (2) longitudinal tracking systems for disease progression monitoring, (3) federated learning for privacy-preserving multi-center training, (4) model interpretability via attention visualization, (5) mobile deployment via TensorFlow Lite, and (6) multilingual speech model extensions.

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